DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Summer Examination – July 2025

Course: B.Tech **Branch: Computer Engineering /Allied** **Semester: VII**

Subject Code & Name: BTCOE705B (Deep Learning)

Max Marks: 60 **Date :25/07/2025** **Duration: 3 Hr.**

Instructions to the Students:

1. *Each question carries 12 marks.*
2. *Question No. 1 will be compulsory and include objective-type questions.*
3. *Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.*
4. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
5. *Use of non-programmable scientific calculators is allowed.*
6. *Assume suitable data wherever necessary and mention it clearly.*

					(Level/ CO)	Marks
Q. 1	**Objective type questions. (Compulsory Question)**					**12**
1	Which of the following is/are Common uses of RNNs?					1
	a. Generate analytic reports	b. Detect fraudulent credit-card transaction	c. Provide a caption for images	d. All of these		
2	. Which of the following is well suited for perceptual tasks?					1
	a. Feed-forward neural networks	b. Recurrent neural networks	c. Convolutional neural networks	d. Reinforce - ment Learning		
3	CNN is mostly used when there is an					1
	a. structured data	b. unstructured data	c. Both a and b	d. None of these		
4	Which of the following is/are Limitations of deep learning?					1
	a. Data labelling	b. Obtain huge training datasets	c. Both a and b	d. None of these		
5	In which of the following applications can we use deep learning to					1

	solve the problem?					
	a. Protein structure prediction	b. Prediction of chemical reactions	c. Detection of exotic particles	d. All of these		
6	Sentiment analysis using Deep Learning is a many-to one prediction task-					1
	a. True	b. False	c.	d.		
7	In CNN, having max pooling always decrease the parameters?					1
	a. True	b. False	c.	d.		
8	Which of the following is a common loss function used for binary classification in deep learning?					1
	a. Mean Absolute Error (MAE)	b. Mean Squared Error (MSE)	c. Binary Cross-Entropy	d. Categorical Cross-Entropy		
9	Which activation function is commonly used in the hidden layers of a deep neural network?					1
	a. ReLU	b. Sigmoid	c. softmax	d.tanh		
10	Which of the following is a common optimization algorithm used in deep learning?					1
	a. Gradient Descent	b. Stochastic Gradient Descent	c. Adam	d.All of these		
11	Which of the following is a common activation function used in the output layer for binary classification?					1
	a. ReLU	b. Sigmoid	c. softmax	d.tanh		
12	Which deep learning technique is used for unsupervised feature learning?					1
	a. Autoencoders	b. CNN	c. RNN	d.Reinforce -ment Learning		
Q. 2	**Solve the following.**					**12**
A)	State and explain key differences between machine learning and					**6**

	deep learning.		
B)	State and explain applications of deep learning.		**6**
Q.3	**Solve the following.**		**12**
A)	Explain the architecture of feed forward neural network with necessary conventions.		**6**
B)	What is back propagation learning? How back propagation learning algorithm is used to update weight vectors?		**6**
Q. 4	**Solve Any Two of the following.**		**12**
A)	Explain architecture of multilayer perceptron.		**6**
B)	Define and explain following terms. (i) Cross Entropy loss (ii) Local Gradient (iii) Back propagated Gradient		**6**
C)	What is Autoencoder? Explain sparse and undercomplete autoencoder in detail.		**6**
Q.5	**Solve Any Two of the following.**		**12**
A)	Which are the different layers of Convolutional Neural Network? Explain their functions.		**12**
B)	What is vanishing gradient problem? What are the challenges in Gradient Descent?		**12**
C)	What is overfitting and underfitting problem? How is it resolved in neural networks?		
Q. 6	**Solve Any Two of the following.**		**12**
A)	What is the need of normalization? Explain different kinds of normalization techniques in neural networks.		**6**
B)	How face recognition process is carried using Deep Learning approach?		**6**
C)	Explain Recurrent Neural network.		**6**
	***** End *****		